

# Git-Vista Security Model

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Status: proposed for V2

Git-Vista is local-first and primarily single-user, but it controls real Git repositories and can execute destructive operations. "Only for me" reduces the identity problem; it does not remove browser-origin attacks, stale tabs, malicious LAN clients, credential leakage, or command races. Repository content is also untrusted input: names, messages, refs and paths are treated as hostile and are validated. Repository code — hooks, filters, and any config key that names an executable — is a different matter. It is designed to run under bounded, irreversible kernel restrictions, tiered by operation kind: Landlock filesystem rules wherever repository code runs, Landlock network rules that scope the network tier to a fixed TCP port list — never to a destination host, see ADR 0028 — seccomp syscall filtering, and, outside the network tier, a `bwrap` namespace boundary. **None of that enforcement is shipped yet.** What is true today: the policy a session will run under is computed and disclosed (ADR 0025), and the pure `argv` chokepoint that will carry it is built and tested (ADR 0017); the shim that actually applies Landlock and seccomp to a hook process does not exist in the tree yet (ADR 0027), so today no hook process is restricted by any of this. It does not claim isolation from a same-uid adversary, and it cannot: see Known Non-Goals.

## Security Objectives

- A website opened in the same browser cannot command the local Git-Vista server.
- A LAN device cannot discover and mutate repositories without explicit pairing.
- No browser or API request can directly supply arbitrary paths, Git `argv`, environment variables, or a shell — enforced at the browser/API surface (ADR 0017). Repository code itself — hooks, filters, configured executables — has its own execution authority once Git invokes it, and is governed separately; see Command Execution and Known Non-Goals.
- A stale or replayed request cannot repeat a mutation silently.
- Provider and Git credentials never enter logs, URLs, API payloads, or browser storage unnecessarily.
- Git-Vista's own read paths cap memory use, output size, and process lifetime — quantified ceilings on what Git-Vista itself spawns, not a blanket bound on the repository. (Implemented for history and file reads: ADR 0022, #63 — every git read goes through `git_vista_git::git_stdout_capped`, which streams under a per-kind cap (8 MiB diff metadata → 413, 200 KB per patch within 5 MB, 2 MB per file) and carries `kill_on_drop`, so a disconnected client kills the child instead of letting it finish into a buffer. History is paged rather than buffered whole, and paging keeps no per-client server state: the entire state of a scroll is one signed offset in the client's cursor, so memory is independent of both repository size and the number of connected clients. The frontend culls to at most 2,000 live rows regardless of camera. The memory bound is a denial-of-service control, not merely a performance measure. Disk use and a hook's own child processes are not bounded by any of this — repository code executes as the

real uid (ADR 0025) with no ceiling of its own, and same-uid resource exhaustion is a Known Non-Goal, not a gap in this control.)

- Recovery data is protected at least as strongly as the repository itself.
- Security remains understandable for one person to operate.

## Assets

- Repository objects, refs, worktrees, index, stashes, and uncommitted changes.
- SSH keys, Git credential-helper material, forge OAuth tokens, and cookies.
- Operation history and recovery refs.
- Repository paths and file contents.
- The ability to push, force-push, create PRs, and modify remote state.

## Trust Boundaries

```
flowchart LR
    B[Browser/PWA]
    T[SSH tunnel or HTTPS]
    S[Git-Vista service]
    G[Git subprocess / gix]
    R[(Allowlisted repositories)]
    C[Credential helpers]
    F[Forge APIs]

    B -- untrusted requests --> T
    T --> S
    S -- typed operations only --> G
    G --> R
    G --> C
    S --> F
```

The browser is not trusted merely because it served the UI. Every mutation goes through authentication, origin validation, repository authorization, generation checks, operation planning, and the per-worktree queue.

## Operating Modes

| Mode                     | Bind                                | Transport                      | Authentication                                                                           | Intended use                                                                                 |
|--------------------------|-------------------------------------|--------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Local                    | 127.0.0.1                           | HTTP<br>localhost              | Launch/session secret +<br>same-origin                                                   | Browser on the Linux/<br>macOS/Windows host                                                  |
| SSH<br>tunnel            | 127.0.0.1 on<br>Linux               | SSH<br>encrypted<br>forwarding | SSH plus Git-Vista<br>session                                                            | Primary iPad-to-Linux<br>workflow                                                            |
| LAN<br>view              | One explicit<br>interface           | Plain HTTP                     | Single-use bootstrap<br>token, view-scoped read-<br>only routes, rate-limited<br>sign-in | Backup path when the<br>SSH tunnel is<br>unavailable, trusted<br>home LAN only (ADR<br>0005) |
| LAN<br>paired,<br>future | Explicit<br>interface               | HTTPS                          | One-time pairing and<br>device session                                                   | Trusted private<br>network without SSH<br>tunnel, full read/write                            |
| Team,<br>future          | Reverse<br>proxy/private<br>network | HTTPS                          | OIDC/passkeys plus<br>RBAC                                                               | Explicit multi-user<br>deployment, not V2<br>default                                         |

Modes are configuration profiles, not a boolean `--public` switch. Each profile has different startup checks and refuses to start when required controls are absent.

## Local and SSH Session Design

Implemented by the session + request-protection layer (M1.04, `git-vista-server:: {session,security}`; ADR 0004). The bootstrap token is written `0600` and delivered via the `gv setup` link's URL fragment (never the server, never a log); everything below is enforced by the `require_auth` layer.

- Generate a high-entropy session secret at every service start unless a durable paired-device session is explicitly configured. (256-bit `getrandom` token per start; the in-memory store means a restart is a full revocation. Durable paired sessions deferred to the LAN/paired milestone.)
- Print/open a bootstrap URL whose one-time secret is exchanged for an `HttpOnly`, `SameSite=Strict` session cookie. Remove secrets from the visible URL immediately. (`gv` prints `.../#s=<token>`; the SPA `POST`s it and strips the fragment with `history.replaceState`. The token is single-use — redeeming it rotates a fresh one — and expires.)
- Require a separate CSRF token on every state-changing request. (Per-session token echoed in `x-git-vista-csrf`, compared constant-time; missing/invalid on a live session is a `403`.)
- Validate `Origin`, `Host`, and content type. Reject `Origin: null` mutations. (Host must be a loopback literal — the anti-DNS-rebinding check; `Origin`, when present, must be same-origin and non-`null`; a present content type on a write must be JSON, blocking form-encoded CSRF.)

- Set a strict Content Security Policy and deny framing with `frame-ancestors 'none'`. (See "Browser Security Headers" — stamped on every response by `security_headers`.)
- Bind only to loopback in Local and SSH modes. (Default bind is `127.0.0.1`; the Host/Origin policy is derived from the bind address.)
- Treat localhost as vulnerable to malicious webpages and DNS rebinding; binding loopback is necessary but not sufficient. (The reason the Host allowlist and session gate exist at all.)
- Expire operation approval tokens quickly and bind them to session, repository, worktree, operation hash, and repository generation. (Session/bootstrap expiry landed here; per-operation approval tokens are a later milestone.)

## LAN View Profile (implemented, ADR 0005)

`gv --lan-view [path]` starts the existing loopback server plus a second listener, bound to one explicit, operator-confirmed LAN IP:

- The second listener serves a structurally reduced router: GET read routes plus `POST/DELETE /api/session` only. Every write, `/api/select`, `/api/rescan`, `/api/clone`, and `/api/delete-clone` route is never registered on it — absence, not a runtime check (`crates/git-vista-server/src/main.rs::api_router`).
- `Host/Origin` on this listener are pinned to the one sanctioned LAN IP:port (`security::HostPolicy::lan`); neither `localhost` nor any other address the machine answers on is accepted, so a DNS-rebinding attempt against the LAN listener fails closed the same way the loopback listener's Host check does.
- Sign-in (`POST /api/session`) on this listener is rate-limited per source IP (`crates/git-vista-server/src/ratelimit.rs`); the loopback listener's sign-in is unaffected.
- Auth is otherwise the same single-use bootstrap-token flow as loopback, sharing one in-memory session store; a session established via either listener carries a `via_lan` flag purely so the UI can hide the Active option — the actual write boundary is the LAN router's absent routes. (Implemented: ADR 0024, #64 — the frontend's CSRF token and `via_lan` flag, formerly two independent `thread_local!`s with no rule tying them together, now live together in one `SessionCore` with a typed `SessionRejection::UiModeChangeWhileLan`, so "a LAN-view session may not select Active mode" is answerable from one place. This is a client-side UI affordance only, reinforcing — not replacing — the write boundary above, which remains the LAN router's absent routes.)
- Accepted, documented risk: plain HTTP means repo contents and the session cookie are readable by anyone on the same network. Suitable for a trusted home LAN, never a guest or shared network — the startup banner and `gv doctor` say so explicitly.
- `gv doctor` and the launch-time exposed-listener kill-check learn the sanctioned second socket: with `--lan-view`, exactly {loopback, the recorded LAN ip} on port 8080 is healthy; anything else is still a SECURITY ERROR that stops the server. Without the flag, behavior is unchanged from M1.05.

## LAN Mode (future, paired HTTPS — write-capable)

The read-only LAN view profile above (ADR 0005) is what's implemented today; this section covers the different, still-future write-capable LAN mode — the "LAN paired" row in the Operating Modes table.

No current write-capable LAN mode exists: the server is hard-limited to `127.0.0.1:8080` plus the read-only LAN view listener above, and the earlier plain-HTTP `--lan` compatibility path was removed. This future mode is a separate paired-HTTPS profile, not a convenience switch, and must:

- Require HTTPS so service workers, credentials, and browser security semantics operate on a secure origin.
- Display a short-lived pairing code locally on the server terminal.
- Pair a browser into a revocable device session; do not use a permanent URL token.
- Show bound interfaces and active paired devices at startup.
- Support revocation and session expiry from the server CLI.
- Rate-limit pairing, authentication, cloning, search, diff, and provider requests.
- Warn that guest Wi-Fi and shared classroom networks are not trusted boundaries.

## Future Team Mode

Do not implement Team mode as "LAN mode with more cookies." It requires:

- External identity through OIDC or passkeys/WebAuthn.
- Per-user repository grants and operation attribution.
- Separate workspaces/worktrees for users who may edit concurrently.
- Administrator-controlled repository roots and provider credentials.
- Audit retention, session revocation, quotas, and reverse-proxy deployment.
- A decision about whether Git commits use service identity or user identity.

These concerns are intentionally outside the V2 local-first implementation.

## Repository Isolation

Implemented by the server-owned catalog (M1.03, `git-vista-server::catalog`; ADR 0003). The catalog is the only path→id resolver and fails closed on anything it did not itself register.

- Configure one or more allowlisted repository roots. (Catalog `AllowedRoots`; the clones root is always allowed, and a server-launched repo allows its own root.)
- Canonicalize discovered paths and reject escapes through symlinks. (Registration canonicalises the repository root and checks component-wise containment, so a `../` traversal or a symlink escaping an allowed root is rejected.)
- Give the browser opaque repository IDs, not arbitrary filesystem paths. (Requests select a worktree by `WorktreeId`; `GET /api/catalog` reports capabilities by id. A malformed id is a `400`, an unknown id a `404`.)

- Resolve git-dir and common-dir through Git/gix for normal, bare, and linked worktree repositories. (`git_vista_git::read_repo_facts` classifies each as bare / main / linked and derives the canonical root.)
- Never follow a browser-provided path to open a file or repository. (No endpoint accepts a path; ids resolve only against the registered set.)
- Report capabilities without exposing absolute paths by default. (Descriptors and the graph label carry only a base name unless `GIT_VISTA_EXPOSE_PATHS` is set.)
- Scope operation stores and recovery refs to the canonical repository identity. (Deferred to M1.09.)
- Treat submodules as separate repositories with explicit opt-in traversal. (Deferred.)
- Never serve `.git` internals as static files.

## Command Execution

- Every Git-Vista-owned spawn site uses direct argv execution; never a shell. (Enforced by a tripwire test: ADR 0017, #144 — `git-vista-server::argv_boundary` scans both native crates and fails on any process-spawn site that is not allowlisted or does not name `git` literally. This governs Git-Vista's own spawn sites only: once invoked, `git` may run a repository hook, filter, or credential helper that itself invokes a shell — that execution is Git's, not a Git-Vista spawn site, and is addressed by Command Execution's hook policy and ADR 0025, not this tripwire.)
- Build argv only from typed operation planners and validated domain values. (Extended to read state: ADR 0022, #63 — a paging cursor is server-authored and HMAC-SHA256 signed. The client may echo it back but may not author it. It is validated in a fixed order — length guard, single dot, bounded base64, constant-time tag comparison, only then JSON parse — so a forged or foreign cursor is rejected before `serde_json` sees attacker-shaped bytes and before any repository walk opens. A cursor scoped to a different repository or worktree returns the generic `400`, deliberately not a distinguishing error, so probing cannot confirm that another target exists.) (Implemented for every served-repository mutation: ADR 0015/0016, #142/#143 — write handlers build a typed `GitOperation`, and `git-vista-server::planner` is the only place a mutating git argv is constructed. Proven at the API boundary by adversarial fixtures: ADR 0017, #144 — no route deserializes a raw command string or argv array, and hostile bodies die at the extractor.)
- Pass `--` where Git supports it and validate full refnames with Git.
- Clear or explicitly set child environment variables. Disable terminal prompts, editors, pagers, and hooks where the operation semantics permit.
- Decide hook policy explicitly. Running repository hooks may execute arbitrary local code; local mode may allow them, but the UI must report that fact and Team mode should default to a restricted policy.
- Apply timeouts, cancellation, stdout/stderr limits, process-group termination, and concurrency quotas.
- Convert raw Git errors into structured safe errors; retain detailed stderr only in local protected logs with credential redaction.

## Sandbox Mechanism Boundaries

The Command Execution and Known Non-Goals sections state what is restricted and disclosed, and — as of this writing — that the enforcing shim is not built yet (ADR 0027). This section states what each mechanism actually covers once it does run, because a boundary that is silently narrower than it sounds is worse than one stated plainly. ADR 0027 (filesystem) and ADR 0028 (network) are the durable record behind each item below.

- **Landlock network rules authorize TCP ports, never hosts.** A single rule granting a port permits `connect()` to that port on every destination; the kernel's rule type carries no address field at all. The network tier's port list (ADR 0028) blocks reaching an arbitrary local port — a stray loopback service, a resolver, this server's own port — and cannot confine which remote host a permitted port reaches.
- **UDP and `AF_UNIX` are not mediated by these Landlock network rules at all.** Only the strict tier's network namespace blocks UDP egress, by removing network access entirely; the network tier's Landlock port rules pass UDP and Unix-domain traffic through unmediated, in either direction.
- **Landlock rules bind resolved inodes, not path strings.** A name excluded from a grant is only actually withheld if the enumeration that builds the grant set resolves symlinks and matches hard-link inodes before deciding what to grant; a granted alias re-opens the excluded file by its own canonical path too. See ADR 0027 for the measured mechanism and the enumerate-and-skip algorithm this requires.
- **A Landlock domain is inherited through `fork` and preserved through `execve`, irreversibly.** This is why a hook or grandchild process stays under the same restriction as its parent — the property the Command Execution and hook-policy sections above depend on.
- **Rule composition differs by scope.** More-permissive nested rules union within one Landlock ruleset (a read-write grant nested under a read-only ancestor adds the extra right); independently applied Landlock domains intersect instead (stacking a second, separate restriction can only narrow what the first already granted, never widen it). Treating the two as interchangeable produces a policy that is wrong in one direction or the other.
- **`/run/docker.sock` is withheld by filesystem policy plus seccomp and descriptor discipline, not by any network rule.** uid 1000's `docker` group membership makes that socket passwordless root once reached; nothing about Landlock's network mediation is what keeps it out of reach.

flowchart TD

```
Q{"Two Landlock<br>restrictions apply"} -->|"same ruleset,<br>nested rule is MORE permissive"| U["UNION —<br>the nested grant adds the extra right"]
Q -->|"separate, independently<br>applied domains"| I["INTERSECT —<br>the second domain can only narrow,<br>never widen, the first"]
U --> Ex1["example: read-write under<br>a read-only $HOME ancestor<br>(ADR 0027)"]
I --> Ex2["example: a hostile hook's<br>own seccomp filter cannot<br>re-widen the launcher's rules"]
```

None of the above is a defect introduced by this document; each is a true property of the underlying kernel mechanism that the sections above did not previously state.

## Remote and Forge Credentials

- Prefer existing Git credential helpers and SSH agents on the Linux host.
- Never ask the browser to upload an SSH private key.
- Store forge tokens server-side using the OS keyring or an encrypted local store.
- Prefer fine-grained, short-lived provider credentials. GitHub recommends GitHub Apps over OAuth apps for tighter permissions and short-lived tokens.
- Request provider scopes per capability and show them before authorization.
- Redact URL userinfo, HTTP authorization, query tokens, and credential-helper output from logs and operation records.
- Make provider logout revoke/delete local credentials.

See [GitHub's app authentication guidance](#).

## Request Integrity

Every mutation request contains:

- Session and CSRF proof.
- Repository and worktree IDs.
- Expected repository generation.
- An idempotency key generated by the client.
- A typed operation body.
- For risky actions, a short-lived approval token from a prior plan.

The server records idempotency results for a bounded period. A retried mobile request returns the original result rather than performing the action twice.

(Generation, hash and expiry are enforced at execution time: ADR 0018, #145 — `planner::validate` refuses a tampered operation hash and an expired plan, and `planner::enforce_fresh` recomputes the repository generation (HEAD, all refs, worktree status) and re-verifies every build-time-held precondition against the live repository immediately before execution. Any drift refuses with a 409 and a client-facing reason — the TOCTOU gap fails closed. Client-supplied generation/idempotency fields arrive with the review roundtrip, M2+.)



## Operation Risk Classes

| Class                | Examples                           | Required control                              |
|----------------------|------------------------------------|-----------------------------------------------|
| Read                 | History, diff, blame               | Authentication, limits, path isolation        |
| Local reversible     | Stage, branch create               | Serialized operation, generation check        |
| History rewrite      | Reset, rebase, amend               | Preview, explicit confirmation, recovery ref  |
| Worktree destructive | Clean, discard, checkout overwrite | Preview, typed file impact, safety checkpoint |
| Remote visible       | Push, PR create/comment            | Explicit target and identity confirmation     |
| Remote destructive   | Force-push, remote branch delete   | Strong warning, lease/CAS, re-auth option     |

No gesture, pressure threshold, swipe, or double-tap directly executes a destructive operation. Touch gestures may select or open a plan; final confirmation is explicit.

## Clone and Network Controls

- Restrict clone schemes by operating mode and configuration.
- Block loopback, link-local, cloud metadata, and private-network destinations when arbitrary URL cloning is exposed beyond local mode, unless explicitly allowlisted.
- Apply clone size/time quotas and stream progress without buffering complete output.
- Store temporary clones under a managed root with ownership metadata and expiry. (Implemented: ADR 0008, #121 — clones persist under `$XDG_DATA_HOME/git-vista/clones`, not a wiped-at-startup temp dir; deletion is explicit via `POST /api/delete-clone`, guarded to paths that canonicalize inside that root.)
- Never delete a clone while an active repository handle references it. (Implemented: ADR 0008, #121 — `delete_clone` refuses the currently open selection with `409`.)
- Treat remote URLs as secrets because they can contain credentials.

## Browser Security Headers

At minimum:

```
Content-Security-Policy: default-src 'self'; object-src 'none'; base-uri 'none';
  frame-ancestors 'none'; form-action 'self'; connect-src 'self'
Cross-Origin-Opener-Policy: same-origin
Cross-Origin-Resource-Policy: same-origin
Referrer-Policy: no-referrer
X-Content-Type-Options: nosniff
Permissions-Policy: camera=(), microphone=(), geolocation=()
Cache-Control: no-store # authenticated API responses
```

Static fingerprinted assets may be immutable. Do not apply `no-store` blindly to the PWA shell and destroy offline startup.

Implemented by `security_headers` (M1.04; ADR 0004), stamped on every response. Two deviations from the baseline above, both documented in ADR 0004: the CSP's `script-src` adds `'wasm-unsafe-eval'` (required by the WebAssembly runtime) and `'unsafe-inline'` (Trunk boots the wasm with an inline module script whose hash changes each build, and the server sets a static header), and `img-src/font-src` are widened to `'self'` `data: /` `'self'` for the inline SVG favicon and the bundled Nerd Font. `Cache-Control: no-store` is applied to API responses only; the SPA shell keeps `no-cache` so offline startup (a later PWA milestone) survives.

## Data at Rest

- Operation history may expose commit messages, branches, and repository paths.
- Default to user-only filesystem permissions on the Linux host.
- Keep browser persistence minimal and clearable.
- Do not cache private diffs/file contents offline by default.
- Recovery refs and safety stashes require retention limits and visible cleanup.
- Provider tokens and device sessions must not be stored in the Git repository.

## Audit and Logging

Log structured events with operation ID, repository ID, worktree ID, risk class, duration, result code, and before/after generation. Do not log request bodies by default. Local logs should be useful without exposing credentials or complete file contents.

## Security Testing

- Route tests for missing/invalid session, CSRF, Origin, Host, content type, and repository access.
- Concurrency tests for clone switching, stale generations, duplicate requests, rebase/reset interleaving, and two browser tabs.
- Property/fuzz tests for ref, path, status, diff, and provider parsers.
- Tests for symlink escapes, linked worktrees, bare repositories, submodules, and repositories with hostile names/messages.
- Output-limit and timeout tests using intentionally noisy/slow child processes.
- Browser tests for cookie and service-worker update behavior.
- Dependency, license, secret, and vulnerability scanning in release CI.

## Known Non-Goals

- Protecting a repository from its own Unix account owner.
- **Fully** sandboxing arbitrary Git hooks in Local mode. Hook policy is **declared and disclosed, not enforced** (ADR 0025) — a session's policy is computed and shown to the user, but no code path suppresses or restricts a hook today, so a `Restricted` session's hooks run exactly like an `Allow` session's would. A same-uid adversary with access to a root-owned daemon socket, or to a writable file some outside process treats as instructions, is out of scope.

- Providing tenant isolation in V2.
- Making remote force-push universally undoable.
- Securing plain HTTP LAN mode; it should not exist as a supported write mode.

## Security Release Gate

No release should claim safe remote or LAN write access until:

- Loopback is the default bind.
- SSH tunnel mode is documented and tested on iPad.
- Every mutation has session, CSRF, origin, generation, and serialization checks.
- Repository paths are allowlisted and opaque to the browser.
- Child processes have time/output limits and credential redaction.
- Recovery behavior is documented for each destructive operation.
- A `SECURITY.md` vulnerability-reporting policy exists.